

Study Project: Adversarial Machine Learning

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Brandenburg University of Technology Cottbus-Senftenberg

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Brandenburg
University of Technology
Cottbus - Senftenberg

General Information (1/3)

- **Lecturer:** *Prof. Dr.-Ing. Andriy Panchenko*
- **Coordinating teaching assistant:** *Asya Mitseva, M.Sc.*
- **Contact:** asya.mitseva@b-tu.de
- **Online Platform:** *Moodle*
 - ▶ Available slots for consultations and Q&A sessions
 - ▶ Announcements, distribution of exercise sheets, etc.
- **ECTS Credits: 8**
 - ▶ Study project: *165 hours $\approx$ 15 hours per week per person*
 - ▶ Self organized studies: *60 hours per person*
 - ▶ Consultation: 1 hour per week per person
- **Assessment**
 - ▶ Graded Q&A session for each exercise sheet
 - ▶ Final presentation of your study project
 - ▶ Final report

- **Master students eligible to register for this course**
 - ▶ Cyber Security Master
 - ▶ Computer Science Master
- Students enrolled in other study programs are **not eligible** to register
 - ▶ eBusiness Master
 - ▶ Artificial Intelligence Master
 - ▶ Künstliche Intelligenz Technologie Master
- **The course is not suitable for first semester Master students**
- **Restrictions for the creation of groups**
 - ▶ Groups of two students, repeating the course
 - ▶ Groups of two students, taking the course for the first time
 - ▶ *Groups of one repeater and one non-repeater currently not possible*

General Information (3/3)

- **New:** The course has a **limited** number of **participants**
 - ▶ Registration deadline has already passed (12 October 2025)
 - ▶ Only officially registered students will be considered during the selection
 - ▶ The test is only one of the selection criteria
- **Advice:** If you do not get place, do **not deregister** immediately
 - ▶ If selected students drop the course, we continue filling the free spots
 - ▶ The process continues until the release of the first practical sheet
- **Request:** Let us know if you decide to drop the course
 - ▶ Via email at asya.mitseva@b-tu.de
 - ▶ Then, other students get the chance to take the course
- Final deregistration of not selected students is handled by us
 - ▶ The deregistration is *not counted* as a failed attempt!

Organization of Study Project (1/4)

- **Lecture period:** from 13.10.2025 to 06.02.2026
 - ▶ All practical sheets and final presentations should be done in this period
 - ▶ No extensions of deadlines possible
- **Working groups:** two students in a single group
- **Exercise sheets**
 - ▶ Five practical exercise sheets
 - ▶ *Biweekly submissions and Q&A sessions* of practical exercise sheets
 - *Continuous assessment: failed submission/Q&A session $\equiv$ failed course*
 - ▶ Final (sixth) exercise sheet for preparing final presentation and report
- **Consultations and Q&A sessions**
 - ▶ Weekly announcement of available time-slots for consultation
 - *General rule: First come, first served!*
 - ▶ *Avoid any (last-minute) requests for consultations via email!*
 - ▶ Predefined time-slot for each group for *Q&A session*
 - *Always take place on Wednesday*

Organization of Study Project (2/4)

- **Deadlines**

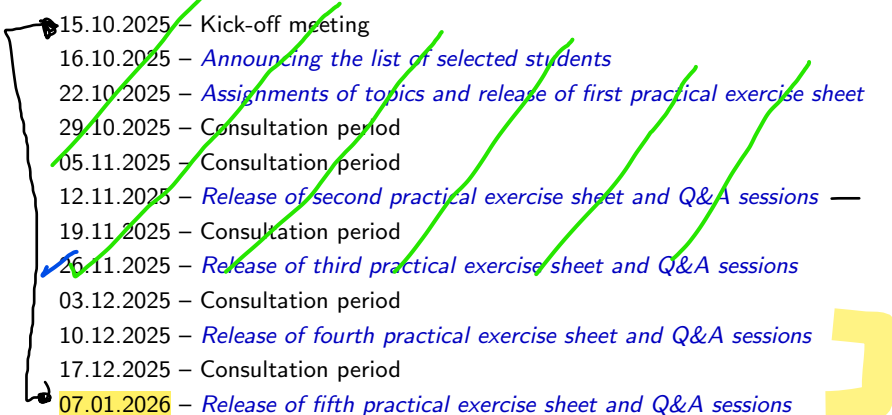
- ▶ Cancellation of course registration possible *before 02.11.2025*
- ▶ Submission of exercise sheet: *biweekly, by Wednesday, 11:59 a.m.*
- ▶ Release of new exercise sheet: *biweekly, Wednesday at 12:00 p.m.*

- **Grading system**

- ▶ Number of *points* for a *single practical sheet*: *maximum 12*
- ▶ Number of *points* for the *final presentation*: *maximum 10*
- ▶ Number of *points* for the *final report*: *maximum 10*
- ▶ *At least 50% of the corresponding points needed to pass assignment*
- ▶ *What if you did not get at least 50%?*
 - In total, you have *two chances for resubmitting* during the whole course
 - *Otherwise, you fail the course*
- ▶ *At least 75% of the total number of points for practical sheets* to be eligible to do final presentation; *otherwise, you fail the course*
- ▶ *At least 70% of the total number of points* for all practical sheets, report, and final presentation to *pass the course*

Organization of Study Project (3/4)

• Overall overview

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- 15.10.2025 – Kick-off meeting
 - 16.10.2025 – *Announcing the list of selected students*
 - 22.10.2025 – *Assignments of topics and release of first practical exercise sheet*
 - 29.10.2025 – Consultation period
 - 05.11.2025 – Consultation period
 - 12.11.2025 – *Release of second practical exercise sheet and Q&A sessions* —
 - 19.11.2025 – Consultation period
 - ✓ 26.11.2025 – *Release of third practical exercise sheet and Q&A sessions*
 - 03.12.2025 – Consultation period
 - 10.12.2025 – *Release of fourth practical exercise sheet and Q&A sessions*
 - 17.12.2025 – Consultation period
 - 07.01.2026 – *Release of fifth practical exercise sheet and Q&A sessions*
 - 14.01.2026 – Consultation period
 - 21.01.2026 – *Release of sixth exercise sheet and Q&A sessions*
 - 28.01.2026 – Consultation period
 - 02.02.2026 – 06.02.2026 – *Final presentations*

Organization of Study Project (4/4)

- **Final presentation and final report**

- ▶ Duration of the presentation: approx. 30 min (excl. project demo)
- ▶ Use of *our L^AT_EX template* for the report is *mandatory*
- ▶ General hints
 - Give clear and concrete problem definition of your project
 - Briefly motivate why the problem is interesting scientific problem
 - Point out the challenges that have to be addressed to solve the problem
 - Provide complete and convincing explanation how you solved the problem and argue why you chose the techniques presented
- ▶ Other additional information will be announced during the semester

- **Last remarks**

- ▶ Try to understand your problem first before start writing source code
- ▶ Try to provide complete and precise solutions
- ▶ Do not try to copy-paste old solutions from previous years
 - We have them as well 😊

Some of Our Topics for Study Project

- **Overall idea**

- ▶ Mount different attacks and/or defenses
- ▶ Using machine learning approaches

- **Some of the possible topics**

- ▶ Attack Detection in Industrial Control Systems (ICS)
- ▶ Generation of Attacks for Industrial Control Systems
- ▶ Implementation and Evaluation of Honeypots
- ▶ Generation of Datasets for Anomaly Detection
- ▶ End-to-end Traffic Correlation Attacks

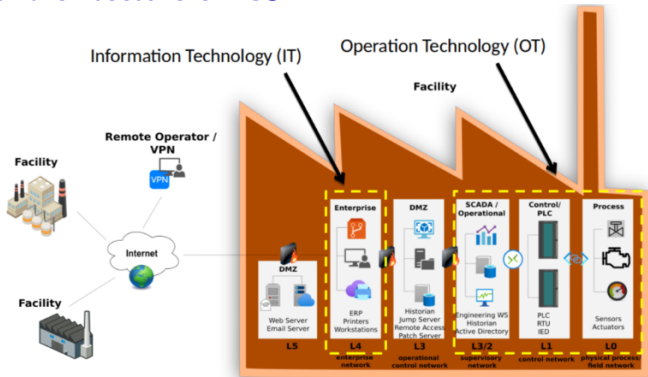
- **Concrete topics may still vary and number of possible topics might be extended**

Generation and Detection of Attacks in ICS (1/3)

- **Industrial Control System (ICS)**

- ▶ *Objective:* Monitor and control various industrial processes
- ▶ *Examples:* Chemical plants, power systems, gas pipelines, etc.

- **General architecture of ICS**



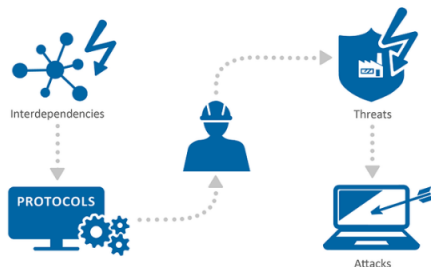
Generation and Detection of Attacks in ICS (2/3)

- **OT is separated from the IT by a demilitarized zone (DMZ)**
 - ▶ No attacks from the IT systems can be launched against OT
- **Current trend in OT systems**
 - ▶ Use of embedded devices with computing capabilities
 - ▶ Use of Ethernet-based protocols
- **Increasing interconnection of OT systems with public networks**
 - ▶ E.g., Internet, mobile communications
- **Problem: New attack surfaces against OT systems**
 - ▶ Communication protocols support (almost) no security functions
 - ▶ (Field) devices are placed outside and are easily accessible to the attacker (e.g., via a wireless connection)

Generation and Detection of Attacks in ICS (3/3)

- Possible projects

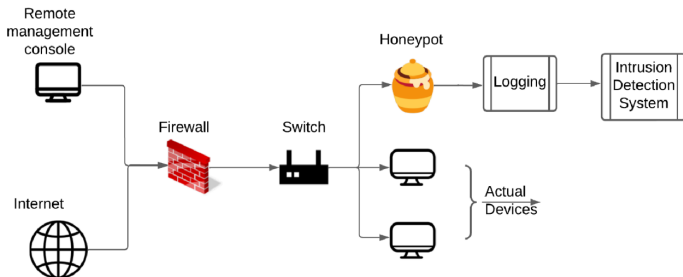
- 1 Explore *different features* and machine learning methods for *detection of ICS attacks* from *network traffic*
- 2 Investigate *different sets of features* extracted from *process data* and different machine learning algorithms in order to *detect attacks in ICS*
- 3 Use of different *machine learning* algorithms to *extend the number of attacks* in existing ICS datasets



Implementation and Evaluation of Honeypots

- **Goal of honeypots:** Decoys to deceive attackers

- ▶ Attract real attackers
- ▶ Record interactions useful for machine learning based attack detection
- ▶ *Advantage:* Early warning about unknown attacks

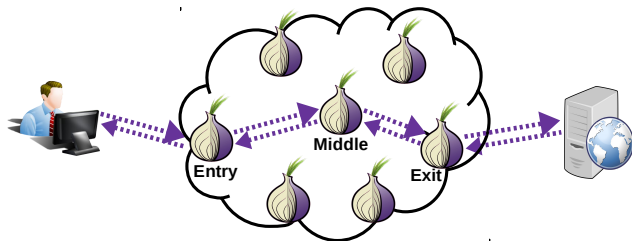


- **Possible projects**

- ▶ Implementation and evaluation of different types of honeypots, e.g.,
 - ① Using deep neural networks or generative AI
 - ② Using open-source network emulators

End-to-end Traffic Correlation (1/2)

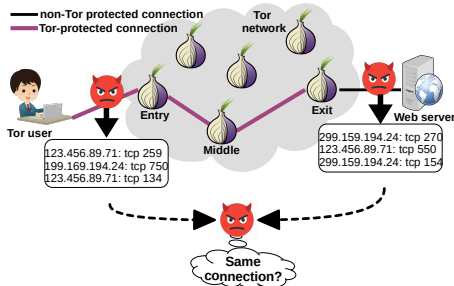
- **Privacy has become a concern**
 - ▶ People leave a large quantity of digital traces on the Internet
- **Encryption cannot fully obscure the content**
- **Solution: Privacy enhancing techniques**, e.g., Tor, VPN
 - ▶ Hide who communicates with whom



- **Problem: Traffic patterns are still visible**
 - ▶ Meta-data such as packet size, direction, and timestamps

End-to-end Traffic Correlation (2/2)

- **Goal: Correlate ingress and egress segments of Tor connection**
 - ▶ Passive observation of traffic patterns
 - ▶ No need to break the encryption



- **Possible projects**
 - 1 Investigate the scalability of different *end-to-end traffic correlation attacks* in different evaluation scenarios
 - ▶ Use of sophisticated simulator for the generation of data
 - 2 Eventually, implementation and evaluation of different *defenses* against end-to-end traffic correlation

Next Steps

- **Announcement of the list of selected students**
 - ▶ If you do not get place, do not deregister immediately
 - ▶ Let us know if you decide to drop the course (see Slide 4)
- **Creation of groups**
 - ▶ On volunteer basis in Moodle by Sunday
 - ▶ Please consider the restrictions, presented in Slide 3
 - ▶ If you cannot find group mate, we will select one for you
- **Topic selection**
 - ▶ Let us know your topic preferences in Moodle on Monday and Tuesday
 - ▶ *However*, final topic assignment is done by us

Questions?